

EXHIBIT J

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLORADO**

**CIVIL ACTION NO. 09-cv-00019-MSK-MEH
United States of America v. CEMEX, Inc.**

**REBUTTAL REPORT
OF
BRADLEY VENNER, PH.D.**



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ON BEHALF OF THE UNITED STATES OF AMERICA

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LIST OF ACRONYMS AND ABBREVIATIONS

AI-539	Kiln O ₂
ARFIMA	Auto-regressive fractionally integrated moving average model
CEMS	Continuous emissions monitoring system
FIC-550	Kiln Feed Rate, tons/hour
FIC-102	Preheater/calcliner coal feed rate, tons/hour
EPA	United States Environmental Protection Agency
LCL	Lower confidence limit with 95% coverage
NO _x	Nitrogen oxides (NO + NO ₂), tons
OE	Oxygen enrichment of tertiary air flow into calciner
P(> t)	P-value, probability of observing a T-value as large or larger than observed if the true value of the parameter was zero.
PSD	Prevention of significant deterioration
T-value	Calculated value of Student's t statistic
TI-550	Kiln Feed Temp
UCL	Upper confidence limit with 95% coverage

I. BACKGROUND

I.1. Information Required by the Federal Rules of Civil Procedure

The following is a list of the items required by the Federal Rules of Civil Procedure:

1. This rebuttal report contains my opinions, conclusions and the reasons therefore;
2. In my deposition, I will provide all materials relied upon and work papers created in connection with this rebuttal report;
3. I received no special compensation for the preparation of this rebuttal report outside of my regular salary received as an Environmental Protection Agency (EPA) employee;
4. A statement of my qualifications and a list of publications I authored within the last ten years are in Appendix A;
5. I have testified twice within the preceding four years as an expert at trial or by deposition, the foregoing proceedings are identified in Appendix A;
6. Appendix B lists the information I considered in forming the opinions expressed in this rebuttal report.

The opinions expressed in this rebuttal report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

I.2. Personal Background

This section of my rebuttal report sets forth portions of my background, education, training, and experience relevant to the substantive issues I address in this report. A more complete CV and a list of publications are attached as Appendix A.

I have more than 15 years of experience in the application of statistics to environmental problems. I received a B.A. in Mathematics from the University of Colorado at Boulder and a Ph.D. in Biostatistics from the Medical University of South Carolina (MUSC). During my graduate studies at MUSC, my emphasis was on environmental risk assessment, and my Ph.D. research involved the application of stochastic models of carcinogenesis to assess the risk of low-dose radiation exposure.

During my time in graduate school, I worked part time at EnSafe, Inc., where I helped with various data and statistical issues related to the cleanup of the Charleston Naval Base. I have been employed by the National Enforcement Investigation Center at EPA (NEIC) since 1998. During my employment, I have had the opportunity to work on a wide range of statistical

issues related to environmental enforcement, such as sampling, risk assessment, source-receptor modeling, and statistical issues related to the enforcement of EPA regulations. During my employment, I have had the opportunity to analyze numerous environmental data sets using regression analysis and related tools. I have worked as an expert on several EPA enforcement cases. A statement of my qualifications, a list documents that I have published in the last 10 years, and my previous trial and deposition testimony in the last 4 years is detailed in Appendix A.

II. SUMMARY OF CONCLUSIONS

- A. My rebuttal of Mr. Hofmann's opinion is restricted to Section 5.3 of his report: "Lyons Expansion Project – Oxygen Enrichment and NO_x Improvements" (Hofmann, p 5-10 to 5-20) and other aspects of his report that depend upon this section. My rebuttal of Mr. Davis' opinion is restricted to Section 4 of his report: "Historical NO_x Emissions Data and Emissions Factors" (Davis, p 16-18) and other aspects of his report that depend upon this section.
- B. Mr. Hofmann claims that "Based on CEMEX's March 1997 Oxygen Enrichment Test (March 1997 test), oxygen enrichment lowers overall NO_x emissions from the pyroprocessing system" and that "these data are more than adequate to assess the impact of OE [oxygen enrichment] on the pyroprocessing system's NO_x generation and facility's overall NO_x emissions with a degree of reliability sufficient for a causation assessment under PSD." (Hofmann, 1-5). In other words, Mr. Hofmann claims that oxygen enrichment causes lower NO_x emissions. Mr. Hofmann offers the statistical analysis presented in Section 5.3 of his report to support this claim. However, this statistical analysis is insufficient to support Mr. Hofmann's claim that the March 1997 test study is sufficient to assess causation. One well-known set of 9 aspects by which an association should be considered for evaluating a claim of causality was provided by Bradford Hill. Very few aspects of a causal relationship are satisfied by Mr. Hofmann's analysis.
- C. The criterion of *specificity* is not satisfied because similar or larger shifts in average NO_x emissions that occurred during oxygen enrichment in the March 1997 Test were observed without changes in oxygen enrichment. One such occurrence was observed after an upset condition occurred during the baseline period; after this upset average NO_x emissions during the test's baseline period declined significantly. This shift in average emission level between the pre- and post-upset baseline is larger than the shift in average emission level between the post-upset baseline and the oxygen enrichment period. Other such occurrences are observed in hourly CEMS monitoring data from 1998. During this time period, oxygen enrichment

was in routine use. A statistical analysis shows that large shifts in the average NO_x emissions as lbs/hr over a 14-day period are a relatively regular occurrence, and an auto-regressive fractionally integrated moving average (ARFIMA) model fit to the same data shows that the data exhibit a long-memory property. These analyses show that changes in the average NO_x emissions over short-term periods are relatively common, while at the same time the process average is relatively stable over longer time periods. Thus, a shift in average emissions over short time periods does not require the assumption that it was caused by a specific process change like oxygen enrichment.

- D. The criterion of *consistency* is not satisfied because a decline in NO_x emissions in the presence of oxygen enrichment is not observed in the CEMEX April 1998 OE test, CEMEX's 1998 NO_x monitoring data and clinker production extracted from the operating record, or CEMEX's historical summary of clinker production and NO_x emissions. Analysis of the NO_x emissions as lbs/ton clinker calculated from CEMEX's historical summaries of clinker production and NO_x emissions shows a clear upward trend in emissions from 1995 to 2001 and that no marked decline is seen after the routine use of oxygen enrichment began in 1997. Analysis of hourly NO_x emissions and oxygen enrichment flow rates compiled from CEMEX operating reports from 1998, a time period in which oxygen enrichment was routinely used, shows that there is a statistically significant increase in NO_x emissions in lbs/hr with the amount of oxygen injected. This is inconsistent with the magnitude and direction of the influence found by Mr. Hoffman.
- E. The criterion of *plausibility* is not satisfied because the changes found by Mr. Hofmann in the relationships between the calciner coal feed, kiln oxygen, and feed rate and NO_x emissions during oxygen enrichment are physically implausible¹. Furthermore, similar reversals in these parameters occur in the post-upset baseline period, further drawing the plausibility of this model into question.
- F. The criterion of *analogy* (consideration of alternate explanations) is not satisfied because Mr. Hofmann does not examine whether other changes in kiln operating conditions could provide a reasonable explanation for his observed changes. Analysis of the relationship between NO_x emissions and other measured values contained in the spreadsheets summarizing the March 1997 test study shows relationships that are much stronger than the relationship with NO_x. These variables may play a confounding role in that they are related to both NO_x emissions and oxygen enrichment. One possible explanation in the way a potential confounding

¹ Walter L. Greer, Expert Report, 1 March 2011; Walter L. Greer, Expert Rebuttal Report, 24 June 2011

variable could be related to the oxygen enrichment time period would be if other kiln operating parameters were changed at the time that oxygen was introduced.

- G. After examining the available evidence as presented in my report, I do not believe that there is sufficient evidence in the March 1997 test to conclude that the association between oxygen enrichment and NO_x emissions proposed by Hofmann and Davis is a causal relationship.
- H. In addition to reviewing the March 1997 test, I analyzed other data sets identified in this report. Based on this examination, it is my opinion that that there is insufficient evidence to conclude that the association between NO_x emissions expressed in lbs/tons of clinker and oxygen enrichment is a causal relationship. There is, however, a statistically significant positive relationship in 1998 data between NO_x emissions expressed as lbs/hr and O₂ enrichment flow rates. Given the lack of a causal relationship between these two variables, the assumption that the emissions factor would not have changed with the introduction of oxygen enrichment would have been the most reasonable position for CEMEX to have taken.

III. OVERVIEW OF STATISTICAL CONCEPTS

III.1. Regression Analysis and Analysis of Variance

Regression analysis is a statistical tool used for understanding the relationship between two or more variables. When the relationship of interest is between two variables, known as the dependent and independent variables, then the regression is said to be single (or univariate). When the relationship of interest is between a single dependent variable and multiple independent variables, the regression is said to be multiple (or multivariate). A detailed discussion of multiple regression and its use in the legal system may be found in a chapter in the Reference Manual on Scientific Evidence (RMSE) entitled *Reference Manual on Multiple Regression*.

Multiple regression analysis is more complicated than single regression analysis but is often necessary for studying real-world phenomena. A single regression analysis may fail to take appropriate variables into consideration, and the omission of these variables can bias substantially the statistical results; that is, create a systematic tendency for an estimate of a model parameter to be too high or too low. According to the *Reference Manual on Multiple Regression*,

An attempt should be made to identify additional known or hypothesized explanatory variables, some of which are measurable and may support alternative substantive hypotheses that can be accounted for by the regression analysis. Thus, in a discrimination case, a measure of the skills of the workers may provide

an alternative explanation – lower salaries may have been the result of inadequate skills (Reference Manual on Scientific Evidence, p. 187)

The term “analysis of variance” has several meanings, and some of these meanings are closely related to regression analysis. Many analysis of variance models can be considered as (multivariate) regression models where one or more independent variables are discrete categories.

III.2. Inferring Causation from Statistical Models

The ability of a statistical analysis to demonstrate causation has been discussed since the founding of the discipline. It is generally accepted that a statistical analysis can only show association, and that the inference that an association is a causal relationship depends upon factors outside a statistical analysis.

One widely known set of criteria to help guide causal inference was developed by Sir Bradford Hill ([Hill, 1965](#)). Hill identified a set of 9 viewpoints (Hill Criteria) by which an association should be examined before “deciding that the most likely interpretation of it is causal.” They were:

1. Strength,
2. Consistency,
3. Specificity,
4. Temporality,
5. Biological gradient,
6. Plausibility,
7. Coherence,
8. Experiment, and
9. Analogy.

Hill was careful to state that aspects were not “hard and fast rules of evidence that must be obeyed before we can accept cause and effect.” Rather,

(w)hat they can do, with greater or less strength, is to help us to make up our minds on the fundamental question – is there any other way of explaining the set of facts before us, is there any other answer equally, or more, likely than cause and effect?

Although the Hill Criteria have been most widely used with epidemiology, they are more universal and can be applied wherever an observed association is interpreted as a causal relationship.

IV. REBUTTAL

Mr. Hofmann claims that “Based on CEMEX’s 1997 test, oxygen enrichment lowers overall NO_x emissions from the pyroprocessing system” and that “these data are more than adequate to assess the impact of OE [oxygen enrichment] on the pyroprocessing system’s NO_x generation and facility’s overall NO_x emissions with a degree of reliability sufficient for a causation assessment under PSD.” (Hofmann, 1-5). In other words, Mr. Hofmann claims that oxygen enrichment causes lower NO_x emissions. Mr. Hofmann offers the statistical analysis presented in Section 5.3 of his report to support this claim. However, in my opinion this statistical analysis is insufficient to support Mr. Hofmann’s claim that the March 1997 test is sufficient to assess causation since, as discussed below, the majority of the Hill Criteria are not satisfied by Mr. Hofmann’s analysis.

All statistical analyses that I present in my report were conducted in the R Statistical Software package (<http://www.r-project.org>). This software is freely available over the internet. Any packages that are not included in the standard release are indicated in the section where they are used.

IV.1. Specificity

Specificity means that an effect always follows a cause. If an effect occurs without the prior occurrence of a supposed cause, this may raise a question as to whether the relationship is causal.

From the viewpoint of specificity, the association identified by Mr. Hofmann between NO_x emissions and oxygen enrichment does not appear causal because a shift in average NO_x emissions as large as observed with the start of oxygen enrichment was observed after an upset condition during the baseline period. Furthermore, shifts of a similar magnitude as observed in the March 1997 test commonly occur over a time period of 14 days in the NO_x monitoring data from 1998, when oxygen enrichment was used on a routine basis.

During the March 1997 test, between 2:00 pm on February 28 and 1:00 am on March 1, there were radical changes in a wide number of control and process variables. These changes included both the calciner and kiln coal feed rates dropping to zero and beginning the natural gas flow rates in both the calciner and kiln. At the same time, many of the process monitoring variables experienced marked shifts, including NO_x emissions.

After this upset, the average level of NO_x emissions decreased in a clearly visible manner (see Figure). I analyzed the post upset decrease by an analysis of variance model for three time

periods within the March 1997 test: pre-upset baseline, post-upset baseline, and post-oxygen enrichment. This analysis is quite similar to the Student's t-test analysis performed by Mr. Hofmann, except that I divided the baseline period into a pre-upset and post-upset segments. The results of this analysis are shown in Figure 1, and the estimated parameters from this analysis are given in Table 1. Note that the change in the average emissions between the pre-upset and post-upset time periods (72.2 lbs/hr) is larger than the change between the post-upset baseline and the oxygen enrichment time periods (45.5 lbs/hr).

Figure 1: Change in average NO_x emissions during March 1997 Test

Solid line: Hourly NO_x emissions, lbs/hr

Dashed line: Average NO_x emissions during time period

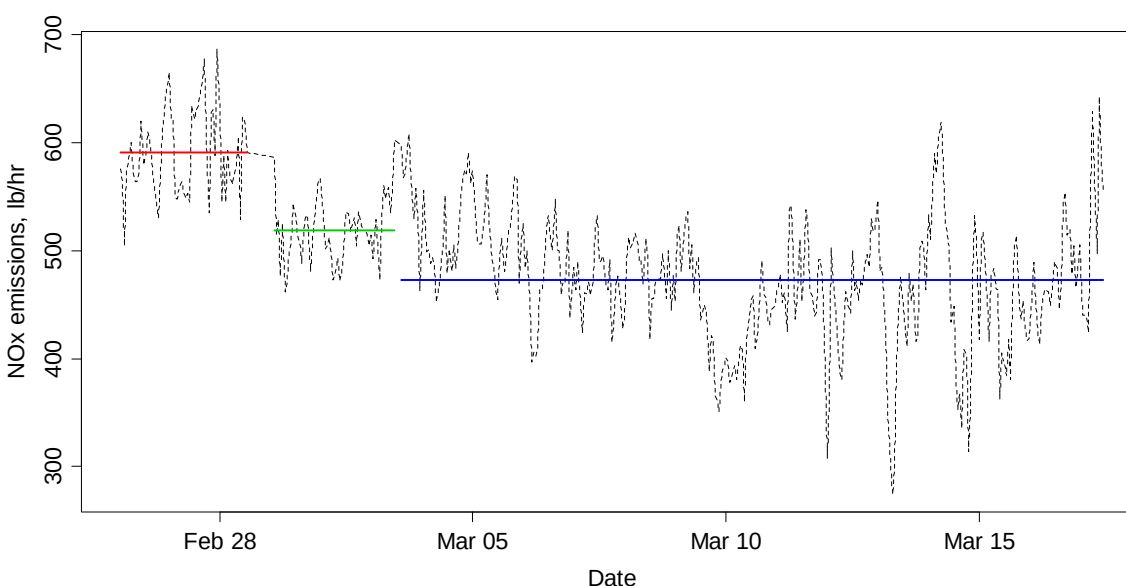


Table 1: Change in average NO_x emissions during baseline and Phase 2 Oxygen Enrichment, March 1997 Test

Parameter	Value	Std. Error	T-value	$P(> t)^2$
Average NO _x emissions, lb/hr, 2/26-2/28, 1997, pre-upset	591.1	6.9	85	$<2 \times 10^{-16}$
Average NO _x emissions, lb/hr, 2/28 – 3/3, 1997, post-upset	518.4	6.9	74	$<2 \times 10^{-16}$
Average NO _x emissions, lb/hr, 3/3 – 3/17, 1997, Phase 2 O ₂ enrichment	472.9	2.9	163	$<2 \times 10^{-16}$

²² $P(>|t|)$ means the probability of the null hypothesis that the parameter is equal to zero is very small.

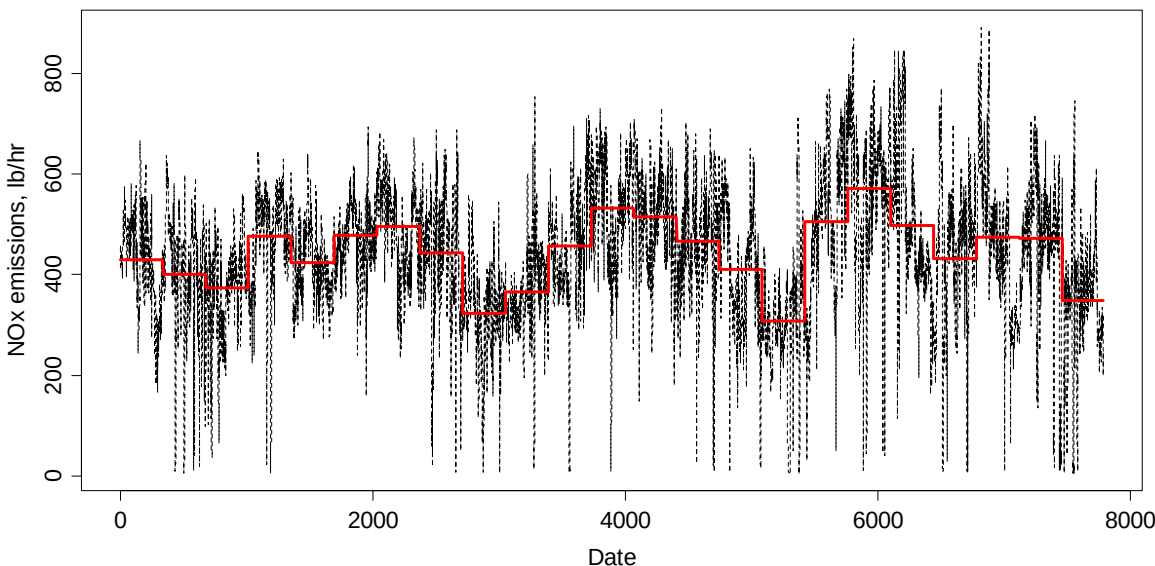
Another way to see the lack of specificity in the proposed association is that similar shifts in the average NO_x emissions, expressed in lbs/hr, are not uncommon over time periods of approximately 14 days in the NO_x monitoring data from the 1998 calendar year, when oxygen enrichment was used on a routine basis. A 14-day period was chosen because this was the approximate duration of the Phase 2 portion of the March 1997 test, where oxygen was injected before the calciner. A subset of the 1998 hourly data was selected when there was both clinker production and NO_x emissions to avoid excessive variability being introduced into the averages due to unrepresentative operation. Data points were excluded that clustered near the x and y axis on a plot of emissions versus clinker. Thus, hourly emissions measurements were excluded if the NO_x emissions were less than 5 or if the clinker production was less than 3.

The data set described above was analyzed by creating a variable that broke the time period into 22 evenly-spaced periods with roughly 336 hourly observations per period. I analyzed the average NO_x emissions as lbs/hr in each 14-day period using an analysis of variance similar to the model previously described; I calculated the shifts between time periods from the differences in the averages within each time period and calculated the probability using the standard propagation of errors method. My analysis shows that large shifts in the average NO_x emissions over a 14-day period are a regular occurrence, even though the average NO_x emissions in lbs/hr appears stationary over this time period. Of the 22 periods I examined, 19 periods contained shifts in NO_x emissions that I identified as statistically significant using the same criteria used by Mr. Hofmann. Three of the shifts observed were larger in magnitude than shifts between the pre-upset baseline and the oxygen enrichment phase of the March 1997 test and 2 were within 20% of the value observed in the March 1997 test. The magnitude of the shifts in NO_x emissions occurring during the year 1998 are illustrated in Figure 2 below, and the estimated parameter values can be found in Appendix C, Table C-4.

Figure 2: Comparison of observed hourly NO_x emissions with piecewise fit of approximately 14 day periods in subset of 1998 data

Solid Line: Estimated average value over approximate 14-day periods.

Dashed line: 1998 NO_x emissions



One reason for the magnitude of these shifts is that the 1998 NO_x emissions data exhibit what is known as the long-memory property. A random process with the long-memory property shows considerable long-term correlation between observations that are widely separated in time. This long-memory property leads to processes that take longer to recover from disturbances than random processes without this property and can lead to relatively persistent shifts in the average level of the process. One statistical model for long-memory processes is the auto-regressive fractionally integrated moving average process (ARFIMA). I fit this model using the routine *fracdiff* from the *fracdiff* package to the NO_x emissions data from 1998 discussed above. The model with the best fit, as measured by Akaike's Information Criteria, was a fractionally integrated model with a single auto-regressive parameter. The values of these parameters can be found in Table 3. I ran a simulation of an ARFIMA process using the estimated parameter values; the estimated auto-correlation function from this simulation is plotted in Figure 3. There is a strong similarity in the estimated and actual auto-correlation function, particularly at longer lags. I also fit the best traditional auto-regressive moving average (ARMA) model to this data; its estimated auto-correlation function is also shown in Figure 3. The traditional ARMA model shows a rapid decay in the auto-correlations at longer lags, which is not observed in the fractionally integrated model.

Figure 3: Comparison of observed and estimated auto-correlation functions from the 1998 NO_x emissions data.

Solid line: sample auto-correlations, NO_x emissions, 1998
Dashed line: Auto-correlations of estimated ARFIMA process
Dotted line: Autocorrelations, best estimated ARMA process

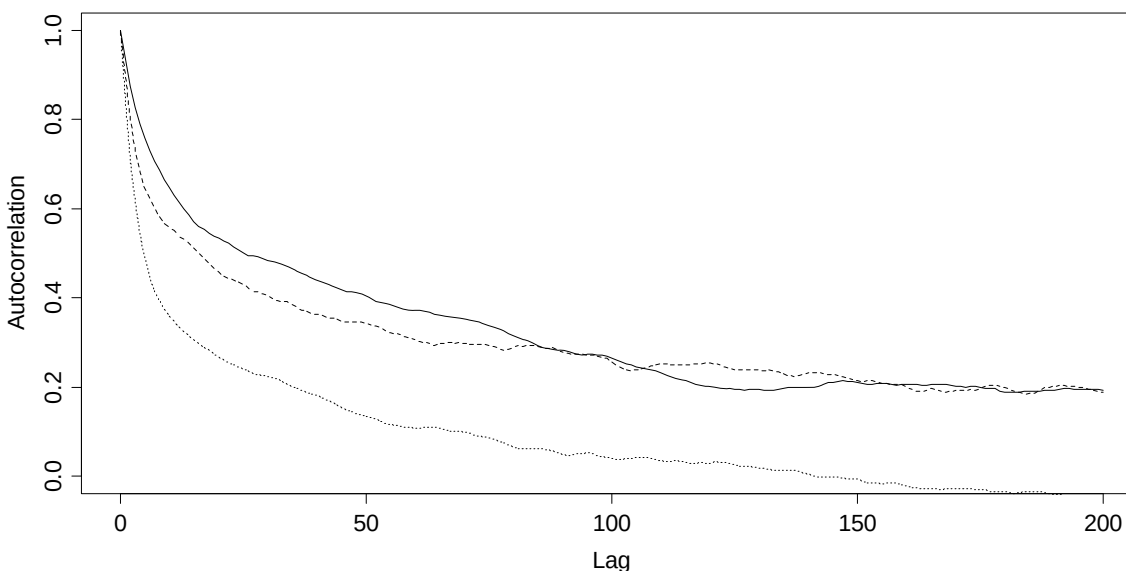


Table 3: ARFIMA parameters estimated from the 1998 NO_x emissions data

Parameter	Estimated Value	Std. Error	T-value	P(> t)
d, fractional integration parameter	0.43	0.02	19	<2 x 10 ⁻¹⁶
ar(1), auto-regression parameter	0.45	0.03	17	<2 x 10 ⁻¹⁶

The long-memory property helps explain the variability in the 14-day average levels of NO_x emissions discussed above. It also shows that time periods longer than 14 days are required to estimate average NO_x emissions values. However, because a fractionally integrated process is stationary, the average NO_x emissions is a well-defined property, and sufficiently long observation periods can estimate this value. Another example of this is that the relative standard deviation of the intercept term in the 14-day time period piecewise linear model was 1.4% relative, indicating very small uncertainty about the average value despite the frequent short-term changes.

IV.2. Consistency

Mr. Hofmann's analysis is inconsistent with four other sets of data, discussed below.

1998 OE Test

Mr. Hofmann misstates the conclusions that can be and were drawn from the April 1998 test when he states that “the 1998 test ... provided CEMEX with support that oxygen enrichment would not result in significant emissions increase.” In fact, the study concluded that there was no consistent relationship between oxygen enrichment and NO_x emissions³, holding all other conditions constant, including clinker production. Thus, this study is inconsistent with the proposed association.

CEMEX NO_x and Clinker Summaries

Other information that does not support a causal interpretation is the historical information of NO_x emissions in lbs/hr and clinker production estimates developed by CEMEX from 1995 to 2001⁴. I calculated NO_x emissions as lbs/ton clinker using these summaries. I then fit a regression line through these data points; the estimated slope parameter is statistically significant. These emissions are plotted in Figure 4 below. This analysis shows that the emissions did not decline after oxygen enrichment was begun in 1997, as would have resulted if oxygen enrichment had the causal relationship proposed by Mr. Hofmann.

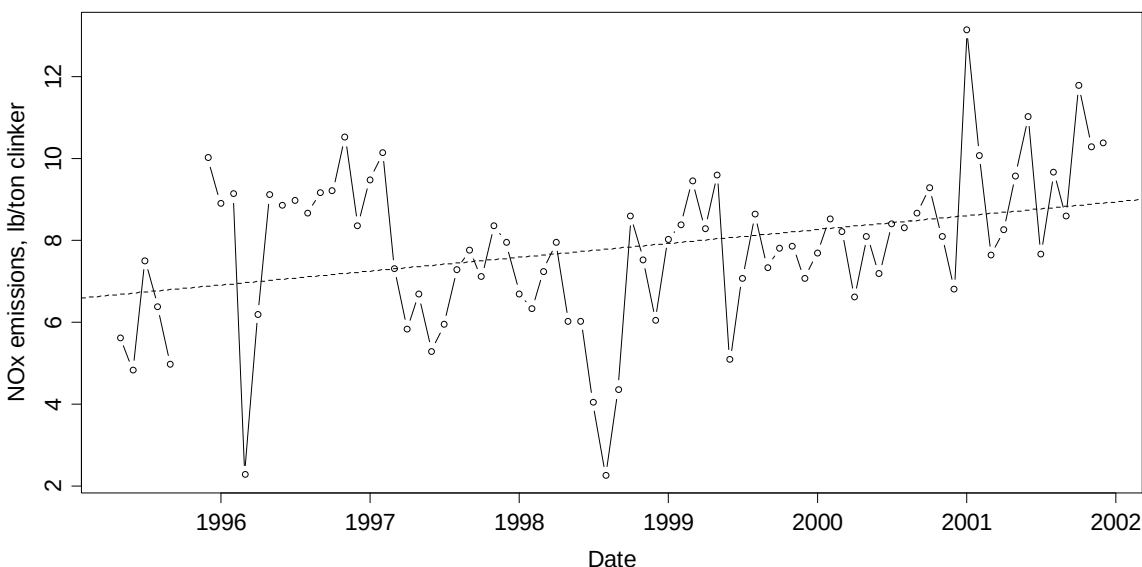
³ The units for NO_x emissions were not given in the report given to CDPHE, but were identified as ppmv by Keith Bauges in his deposition discussing Exhibit 5 (Deposition of Keith Bauges, July 30, 2010, p. 164-167).

⁴ Monthly clinker production for 1995 to 2005 is from CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849-628852. Monthly NO_x emissions for 1995 to 2005 are from CEMEX, NO_x Emissions – Kiln – Historical Summary, CMXLYO00628845-628848.

**Figure 4: Calculated monthly NO_x emissions in lbs/ton from historical CEMEX data
May 1995 to Dec 2001**

Solid line: Calculated monthly NO_x emissions

Dashed line: Increase in monthly NO_x emissions over time from linear regression



**Table 4: Parameters from linear regression to calculated monthly NO_x emissions
in lbs/ton clinker, May 1995 to Dec 2001**

Parameter	Value	Std. Error	T-value	P(> t)
Intercept, NO _x emissions, lb/ton clinker, May, 1995	6.68	0.42	16	<2 x 10 ⁻¹⁶
Slope, NO _x emissions, lb/ton clinker, Increase/Year	0.338	0.11	3.1	0.002

1998 Hourly NO_x CEMS and Clinker Data

Another analysis that is not consistent with a causal interpretation of the proposed association is the estimated relationship between oxygen enrichment and NO_x emissions in 1998, when oxygen enrichment was used routinely. I fit a simple regression model between the NO_x emissions calculated as lbs/ton clinker and oxygen enrichment amounts to the subset of the 1998 data described previously. This analysis showed no statistically significant decline in NO_x emissions with the amount of oxygen injected. The parameters for this model are reported in Table 5, below. A plot of the NO_x emissions versus oxygen enrichment rates, as well as the estimated regression line, are shown in Figure 5.

This analysis shows an increase in the NO_x emissions expressed as pounds/hour in relationship to the O₂ enrichment flow rates; this slope is statistically significant. In contrast to the finding by Mr. Hofmann of a slope of -0.001 (Hofmann, p 5-12 and Appendix D), the

relationship in the 1998 data is 0.0016. Thus, the magnitude of the relationship is the approximately same but the direction of the effect is in the opposite direction. Thus, the association from this data set is inconsistent with the association from the March 1997 test.

Figure 5: Relationship between oxygen enrichment flow rates and hourly NO_x emissions, 1998

Hollow circles: Hourly NO_x emissions, lb/ton clinker, 1998

Dashed line: Estimated linear relationship between NO_x emissions and O₂ enrichment

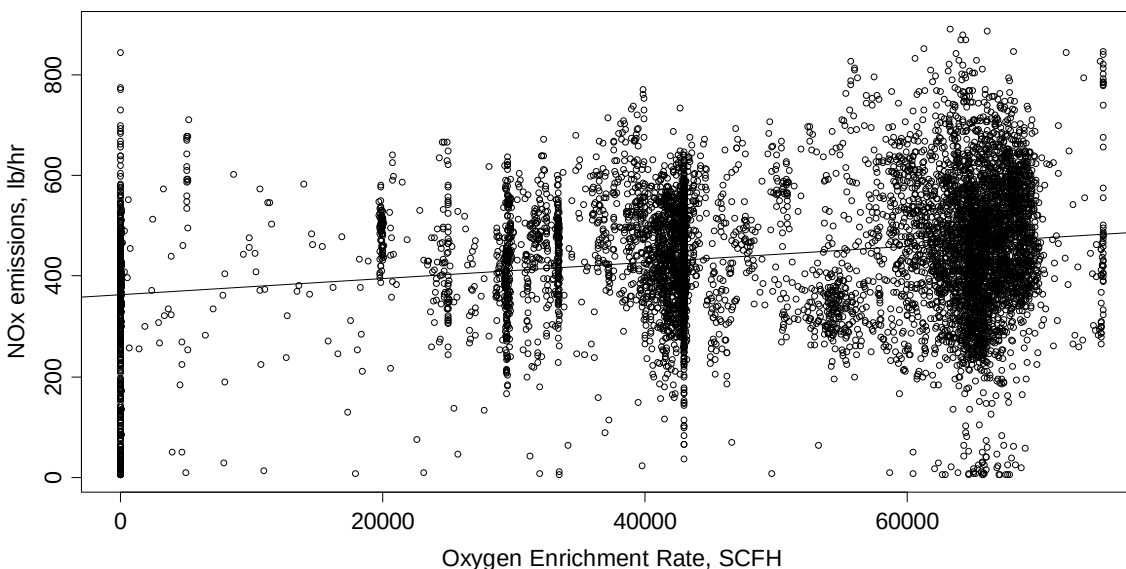


Table 5: Parameters from linear regression fit to oxygen enrichment flow rates and hourly NO_x emissions, 1998

Parameter	Estimate	Std. Error	T-value	P(> t)
Intercept, NO _x emissions, lb/ton clinker, No O ₂ enrichment	363	4	88	<2 x 10 ⁻¹⁶
Slope, NO _x emissions, lb/ton clinker, per O ₂ , scfh	.0016	7.7 x 10 ⁻⁵	21	<2 x 10 ⁻¹⁶

IV.3. Plausibility

From the viewpoint of plausibility, the proposed association does not appear causal because Mr. Hofmann's changes in the relationships between NO_x and calciner coal feed, kiln oxygen, and feed rate and NO_x emissions are physically implausible⁵. Furthermore, similar shifts in these parameters occur in the post-upset baseline period, further drawing the plausibility of this model into question.

⁵ See Walter L. Greer, Expert Report, 1 March 2011; Walter L. Greer, Expert Rebuttal Report, 24 June 2011.

Mr. Hofmann presents an analysis in this report that he terms “analysis of variance”. The purpose of this analysis was to investigate “the possible dependence of NO_x emissions on other operating parameters when oxygen is and is not injected into the tertiary air duct.” (Hofmann, p 5-16). The analysis presented in Mr. Hofmann’s report is not truly an ANOVA analysis, but is a series of univariate regression models between NO_x emissions and three operating parameters, which he identifies as “Kiln O₂ concentration (%),” “coal feed to the calciner (tons per hour),” and “feed rate (tons per hour).” These models are fit to data from three different time periods (pre-oxygen enrichment, post-oxygen enrichment, and all data). Mr. Hofmann does not explicitly identify which variables from the spreadsheet summarizing the March 1997 test were used, but I was able to reproduce his analysis using FIC-102 as “coal feed to the calciner,” FIC-550 as “feed rate (tons per hour) and AI-539 as “Kiln O₂ concentration (%).”

Mr. Hofmann’s analysis estimates different values of the slope parameter in the linear relationship between these parameters and NO_x emissions as lbs/hr in the study periods before and after oxygen enrichment. Mr. Hofmann shows that for one of these parameters, the coal feed rate, there is a different slope in the pre-OE baseline. Mr. Hofmann concludes that “Based on these assessments, it appears likely OE has at least some beneficial effect in reducing the amount of NO_x generated in the pyroprocessing system.” (Hofmann, p 5-16). However, another interpretation is that this analysis casts doubt on the causal role of oxygen enrichment. Mr. Hofmann’s assertion is that when oxygen is being injected, the fundamental relationships between NO_x and calciner coal feed rate changes from a positive to a negative relationship. Mr. Hofmann’s model implies that while oxygen is being injected, any additional amount of coal burned in the calciner causes a decrease in the amount of NO_x generated. This is physically implausible as a causal assertion⁶. It is more likely that this relationship is an artifact rather than a meaningful relationship.

The plausibility of Mr. Hofman’s assertion is brought into greater doubt when the change in the NO_x emissions in the post-upset baseline conditions is considered. I have re-analyzed the relationships posited by Mr. Hofmann but with the baseline “NO OE” condition separated into a pre-upset and post-upset period, as previously discussed in this report. The results of these analyses can be found in Appendix C, Table C-1, C-2 and C-3. In these analyses, the reversal in the relationship between coal feed rate observed in Mr. Hofmann’s analysis also appears in the post-upset baseline and the relationship is quite similar in the post-upset baseline and the oxygen enrichment periods. Furthermore, the relationship between kiln feed rate and NO_x, which was found to be consistently negative by Mr. Hofmann, now reverses – there is a positive slope during the pre-upset baseline period and a negative slope in the post-upset and oxygen

⁶ See Walter L. Greer, Expert Report, 1 March 2011; Walter L. Greer, Expert Rebuttal Report, 24 June 2011.

enrichment periods. These re-analyses cast serious doubt on the causal inference made by Mr. Hofmann. Instead, the changes in these parameters are more likely to be mere statistical associations than causal relationships.

IV.4. Analogy

One way to understand the viewpoint of analogy is to examine the causal relationship that is proposed and to compare it to other possible explanations for the observed association. From this viewpoint, the proposed association does not appear to be causal because other variables have a much stronger relationship with NO_x emissions than oxygen enrichment. Mr. Hofmann did not examine whether other changes in kiln operating conditions could provide a reasonable explanation for the observed changes in average NO_x emissions levels. One such of the many possible analyses that could be undertaken is proposed below.

The March 1997 test was not designed to study the relationship between NO_x emissions and oxygen enrichment, but was focused on the effects of oxygen enrichment on clinker production. As a result, there was not explicit control of all operating parameters that could affect NO_x emissions. Thus, another explanation for the proposed association is that the change in NO_x emissions was caused by changes in operating conditions that were associated with the test conditions.

Although Mr. Hofmann did examine the effects of three operating parameters, a simple correlation analysis shows that the 3 variables identified by Hoffman do not have the largest correlations with NO_x emissions as lbs/hr. Instead, the single variable with the greatest relationship to NO_x generation in the March 1997 test data is “Kiln Feed Temp,” which is the column labeled TI-550 in the spreadsheet, as determined by an automated variable selection routine *regsubsets* from the *leaps* package as implemented in the R programming language. The relationship between this variable and NO_x emissions is visibly non-linear, and in a quadratic regression model the quadratic term was statistically significant. The parameters for this model are given in Table 6 and the relationship is illustrated in Figure 6 below. As can be seen in this figure, as the kiln feed temperature increases, the emissions decrease. This non-linear relationship holds across the pre-upset baseline, post upset baseline, and Phase 2 oxygen enrichment time periods. As can be seen from the graph, there are more datapoints from the pre-upset baseline at lower temperatures than during oxygen enrichment.

Figure 6: Relationship between Kiln Feed Temp and NO_x emissions, March 1997 Test
Circles, Triangles, Crosses: Observed data in pre-upset, post-upset baseline and O₂ enrichment
Solid line: Estimated quadratic relationship between kiln feed temperature
and NO_x emissions, lb/ton clinker

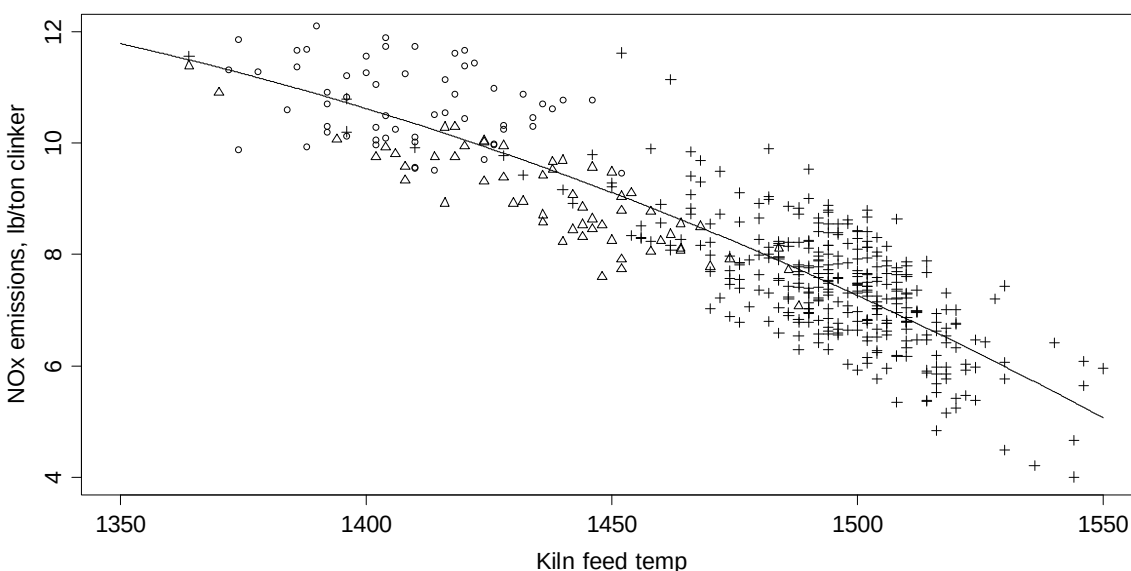


Table 6: Parameters from quadratic fit between TI-550 and NO_x emissions as lb/ton clinker, March 1997 Test

Parameter	Estimate	Std. Error	T-value	P(> t)
NO _x emissions, lb/ton clinker, Kiln Feed Temp = 1364	11.5	0.2	99	<2 x 10 ⁻¹⁶
NO _x emissions, lb/ton clinker, per degree, Kiln Feed Temp	-0.02	4.3 x 10 ⁻³	-5.1	6 x 10 ⁻⁷
NO _x emissions, lb/ton clinker, per degree squared, Kiln Feed Temp	6.8 x 10 ⁻³	2.2 x 10 ⁻³	-3.1	2.2 x 10 ⁻³

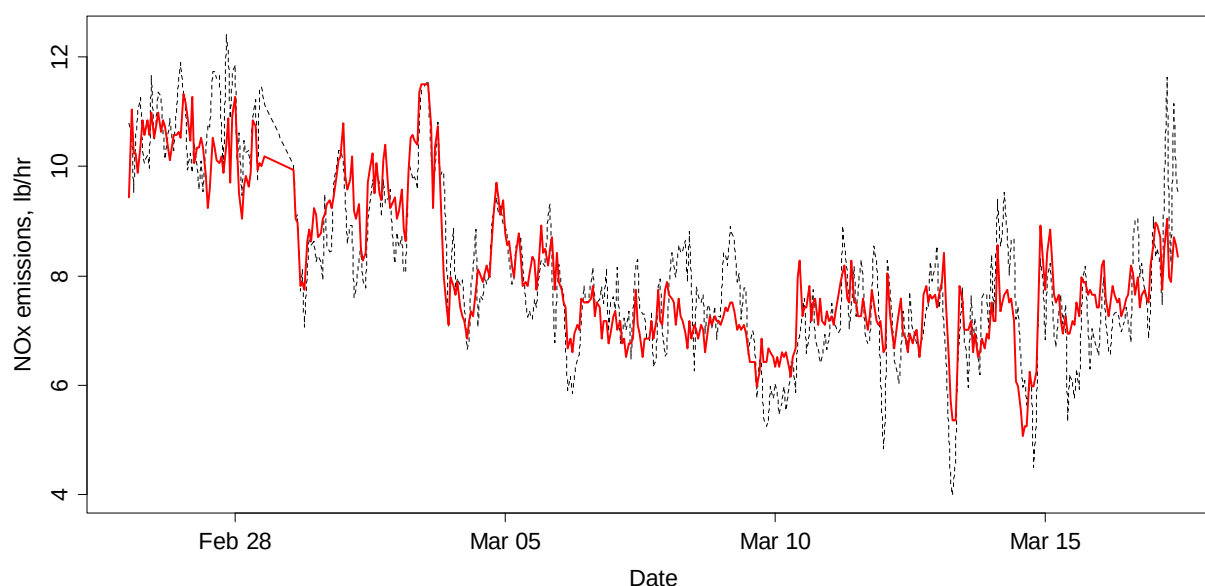
Another view of the kiln feed temperature/NO_x relationship is to compare the estimated values from the regression model in Figure 6 to the observed values of NO_x emissions over time. This viewpoint is illustrated in Figure 7 below. This figure shows that the overall relationship between NO_x emissions and kiln feed temperature is relatively consistent throughout the time period of the study, including the pre-upset and post-upset baseline periods. Thus, no parameter reversal, which implies a change in the fundamental relationship between the parameters, is necessary to describe the changes in emissions in time, unlike the models proposed by Mr. Hofmann. If an additional parameter is added to the model related to the rate of oxygen enrichment, the relationship is not significant when the NO_x emissions are expressed in terms of

lbs/ton of clinker and is positive when NO_x emissions are expressed in terms of lbs/hr. The latter positive relationship holds whether oxygen enrichment is expressed as the amount of oxygen enrichment flow rate in standard cubic feet per minute or simply whether oxygen is injected at all. The parameters for these models are given in Appendix C, Table C-5. The fact that oxygen enrichment has either no relationship or a positive relationship implies that the changes in kiln feed temperature may not be related to changes in oxygen enrichment but are due to some other cause.

Figure 7: Relationship between Kiln Feed Temp and NO_x emissions, lb/ton clinker, over time

Solid line: Estimated NO_x emissions based on quadratic regression with kiln feed temp

Dashed line: Observed NO_x emissions, lb/ton clinker



The fact that there is a strong association between kiln feed temperature and NO_x emissions does not imply that there is a causal relationship between these variables, of course. While the kiln feed temperature fits all the statistical requirements of a confounding variable, whether it fits the causal relationship required to be a true confounder would require a similar examination as has been conducted in this report for the association between NO_x emissions and oxygen enrichment.

IV.5. Strength

The statistical analysis conducted by Mr. Hofmann depends upon the assumption of statistically independent observations. As shown in the previous section, the NO_x emissions data from 1998 exhibit the long-memory property, which is a very strong form of statistical dependence. Thus, the strength of the association demonstrated by the use of the t-test used by

Mr. Hofmann may be invalid if the 1997 emissions data also exhibit the long-memory property. Thus, Mr. Hofmann has not developed a statistically valid demonstration of the strength of his proposed association.

IV.6. Biological Gradient

Although the viewpoint of biological gradient may not seem to apply to a study from a cement kiln, one way to understand this viewpoint is that it refers to whether the magnitude of the effect increases as the magnitude of the cause increases. From this viewpoint, at a superficial level the association proposed by Mr. Hofmann does appear causal, as Mr. Hofmann presents a statistically significant decline in NO_x emissions with O₂ enrichment rates (slope = -0.001) (Hofmann, p 5-15 and Appendix D). However, if this analysis is restricted only to the oxygen enrichment time period, this relationship reverses and becomes positive with a slope of .0008, which is the same magnitude but opposite in direction. Thus, from this viewpoint, the evidence is ambiguous at best.

IV.7. Experiment

Although from this viewpoint, it may appear that the evidence for causation is strong, the 1997 test was not designed to be an experiment on the relationship between NO_x emissions and oxygen enrichment, and instead was intended to study the relationship between oxygen enrichment and clinker production. This fact is explored in greater detail in the expert report of Dr. Fox. Furthermore, although the 1998 test had a number of flaws in both design and execution, as discussed by Mr. Greer, it was intended to study this relationship and came to a different conclusion than Mr. Hofman. Thus, the experimental evidence is ambiguous at best.

IV.8. Coherence

I did not examine the proposed association from the viewpoint of coherence, since this requires specific expertise in the operation of cement kilns. However, my colleague Mr. Greer has examined the arguments provided by Mr. Hofmann, Dr. Davis, and Dr. Young as to their belief as to why oxygen enrichment would cause a decline in NO_x emissions and has found serious flaws in these arguments⁷. Thus, the evidence for coherence is ambiguous at best.

IV.9. Temporality

The viewpoint of temporality is whether the cause precedes the effect. This viewpoint is closely related to specificity. As previously discussed, although the introduction of oxygen

⁷ Walter L. Greer, Expert Report, 1 March 2011; Walter L. Greer, Expert Rebuttal Report, 24 June 2011.

enrichment did proceed a decline in the average level, the average level decreased by an even larger amount in the post-upset baseline. Thus, from this viewpoint, the evidence for causation is ambiguous at best.

IV.10. Conclusion

Dr. Hill concluded his discussion on the viewpoints by which to view an association as potentially causal by stating:

What they can do, with greater or less strength, is to help us to make up our minds on the fundamental question – is there any other way of explaining the set of facts before us, is there any other answer equally, or more, likely than cause and effect? (Hill)

After examining the available evidence as presented in my report, I do not believe that there is sufficient evidence in the March 1997 test to conclude that the association between oxygen enrichment and NO_x emissions proposed by Hofmann and Davis is a causal relationship.

In addition to reviewing the 1997 test, I analyzed other data sets identified in this report. Based on this examination, it is my opinion that that there is insufficient evidence to conclude that the association between NO_x emissions expressed in lbs/tons of clinker and oxygen enrichment is a causal relationship. Given the lack of a causal relationship between these two variables, an assumption that the emissions factor would not have changed with the introduction of oxygen enrichment would have been the most reasonable position for CEMEX to have taken.

APPENDIX A: LIST OF PUBLISHED DOCUMENTS, PRESENTATIONS, AND PREVIOUS LITIGATION EXPERIENCE

Education:

Ph.D., Biostatistics, 2001	Medical University of South Carolina
B.A., Mathematics, 1987	University of Colorado at Boulder

Employment History:

U.S. Environmental Protection Agency National Enforcement Investigations Center (NEIC)	1998 - present
EnSafe, Inc. Environmental Scientist	1993 - 1998

Oral Presentations:

Joint Statistical Meeting, Denver, CO, Informative priors for computing the UCL from a set of asbestos measurements with low total counts. August, 2008.

Pittcon, Chicago, IL. Performance Approach: Evolution or Revolution. February 2007.

National Conference on Managing Quality Systems; New Orleans, LA. A Bayesian approach to measurement uncertainty for laboratory accreditation, April 2003.

Waste Testing and Quality Assurance Conference; Washington, DC. The impact of subsampling variability on TCLP measurements, August 2002.

Publications:

E. Betterton, J. Lowry, R. Ingamells, B. Venner. Kinetics and Mechanism of the Reaction of Hypochlorite with Azide. Journal of Hazardous Materials 182 (2010) 716-722.

R. W. Gerlach, J.M. Nocerino, C. A. Ramsey and B. Venner. Gy sampling theory in environmental studies 2. Subsampling error estimates. Analytica Chimica Acta 490 (2003) 159-168

Litigation Experience:

Deposition taken, United States v Louisiana Generating, LLC, 2011 (CIVIL ACTION NO. 09-100-RET-CN)

Deposition taken, United States v Alabama Power Company, 2010 (Civil Action No. 2:01-CV-00152-VEH, N.D. Alabama)

Technical consultation, United States v. W.R. Grace, Missoula, MT 2007

APPENDIX B: DOCUMENTS CONSIDERED

DEPOSITIONS

Kevin C. Kelley, Oral 30(B)(6) Deposition Of Representative Of CEMEX, Inc., December 16, 2010

Keith Baugues, Deposition taken July 30, 2010

PAPERS

Reference Manual on Scientific Evidence, Second Edition. Federal Judicial Center, 2000. Accessed at [http://www.fjc.gov/public/pdf.nsf/lookup/sciman00.pdf/\\$file/sciman00.pdf](http://www.fjc.gov/public/pdf.nsf/lookup/sciman00.pdf/$file/sciman00.pdf)

Long Range Dependence, Gennady Samorodnitsky, Foundations and Trends in Stochastic Systems, Vol. 1, No. 3 (2006) 163–257. Accessed at <http://www.orie.cornell.edu/~gennady/LRD-NOW.pdf>

Austin Bradford Hill, “The Environment and Disease: Association or Causation?”, Proceedings of the Royal Society of Medicine, 58 (1965), 295-300. Accessed at <http://www.edwardtufte.com/tufte/hill>

REPORTS

Expert Report of Gerald L. Young, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011.

Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011.

Expert Report of Kevin Davis, Ph.D, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011.

Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011.

Rebuttal Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011.

Expert Report of Phyllis Fox, Ph.D, PE, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011.

Rebuttal Expert Report of Phyllis Fox, Ph.D, PE, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011.

DATA SOURCES

CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO000628849-628852

CEMEX, NO_x Emissions – Kiln – Historical Summary, CMXLYO000628845-628848.

CMXLYO 7314 - 7437

CMXLYO	7595 - 7717
CMXLYO	7840 - 7963
CMXLYO	8122 - 8241
CMXLYO	8394 - 8456
CMXLYO	8709 - 8765
CMXLYO	8936 - 9059
CMXLYO	10045 - 10168
CMXLYO	10326 - 10385
CMXLYO	10403 - 10462
CMXLYO	10615 - 10739
CMXLYO	10866 - 10925
CMXLYO	11108 - 11231
CMXLYO	11356 - 11358
CMXLYO	3624 - 3655
CMXLYO	3694 - 3723
CMXLYO	3763 - 3822
CMXLYO	4019 - 4080
CMXLYO	4284 - 4345
CMXLYO	4605 - 4634
CMXLYO	4791 - 4852
CMXLYO	5059 - 5119
CMXLYO	9269 - 9330
CMXLYO	9519 - 9574
CMXLYO	9743 - 9748
CMXLYO	9758 - 9819
CMXLYO	206613 - ?*

APPENDIX C: TABLES OF PARAMETER VALUES

Table C-1: Re-analysis of Mr. Hofmann's "ANOVA" relationship between NOx and the calciner coal rate (FIC-102)				
Parameter	Estimate	Std. Error	T-value	P(> t)
Average emissions during pre-upset baseline if feed rate was zero	-12.5	367	-.034	.97
Emissions per Coal Feed Rate, tons/hour during pre-upset baseline	96.6	59	1.6	.10
Average emissions during post-upset baseline if feed rate was zero	798	288	2.7	.005
Emissions per Coal Feed Rate, tons/hour during pre-upset baseline	-45.6	75	-1.9	.06
Average emissions during post-upset baseline if feed rate was zero	729	729	15.2	$<2 \times 10^{-16}$
Emissions per Coal Feed Rate, tons/hour during oxygen enrichment	-35.5	59	-2.2	0.3
Table C-2: Re-analysis of Mr. Hofmann's "ANOVA" relationship between NOx and the kiln feed rate (FIC-550)				
Parameter	Estimate	Std. Error	T-value	P(> t)
Average emissions during pre-upset baseline if feed rate was zero	310	164	1.9	.06
Emissions per Kiln Feed Rate, tons/hour during pre-upset baseline	3.2	1.9	1.7	.09
Average emissions during post-upset baseline if feed rate was zero	693	105	6.5	1×10^{-10}
Emissions per Kiln Feed Rate, tons/hour during pre-upset baseline	-1.9	2.2	-2.1	.02

Average emissions during oxygen enrichment if feed rate was zero	733	49	15.1	$<2 \times 10^{-16}$
Emissions per Kiln Feed Rate, tons/hour during oxygen enrichment	-2.6	2.0	-3.0	0.003

Table C-3: Re-analysis of Mr. Hofmann's "ANOVA" relationship between NO_x and the kiln O₂ level (AI-539)

Parameter	Estimate	Std. Error	T-value	P(> t)
Average emissions during pre-upset baseline if O ₂ was zero	454	51	1.9	.06
Emissions per O ₂ , percent during pre-upset baseline	59.3	22	2.7	.007
Average emissions during post-upset baseline if O ₂ was zero	343	99	3.5	6×10^{-4}
Emissions per O ₂ , percent during post-upset baseline	75	48	-.35	.73
Average emissions during oxygen enrichment if kiln O ₂ was zero	433	16	26	$<2 \times 10^{-16}$
Emissions per O ₂ , percent during oxygen enrichment	17	23	-1.7	0.08

Table C-4: Changes in average emissions over approximate 14-day periods, 1998 Hourly Monitoring Data

Parameter	Value	Std. Error	T-value	P(> t)
Average Value	430	6.3	83.5	$<2 \times 10^{-16}$
Shift 1	-29	13	-2.3	1.2×10^{-2}
Shift 2	-27	13	-2.2	1.5×10^{-2}
Shift 3	101	13	8.0	4×10^{-7}
Shift 4	-50	13	-4.0	3.3×10^{-5}
Shift 5	53	13	4.1	1.7×10^{-5}
Shift 6	19	13	1.5	6.9×10^{-2}
Shift 7	-53	13	-4.2	1.6×10^{-5}
Shift 8	-119	13	-9.4	$<2 \times 10^{-16}$
Shift 9	43	13	3.4	3.7×10^{-4}

Shift 10	90	13	7.1	6.4×10^{-13}
Shift 11	75	13	5.9	1.5×10^{-9}
Shift 12	-17	13	-1.3	8.9×10^{-2}
Shift 13	-49	13	-3.9	4.9×10^{-5}
Shift 14	-56	13	-4.4	4.6×10^{-6}
Shift 15	-102	13	-8.1	4.1×10^{-16}
Shift 16	198	13	15.6	$<2 \times 10^{-16}$
Shift 17	66	13	5.2	1.1×10^{-7}
Shift 18	-73	13	-5.7	5.1×10^{-9}
Shift 19	-66	13	-5.2	9.9×10^{-8}
Shift 20	43	13	3.4	3.6×10^{-4}
Shift 21	-2	13	-0.2	0.4
Shift 22	-123	13	-9.7	$<2 \times 10^{-16}$

Table C-5: Relationship between NO_x, TI-550 and O₂, March 1997 Test

Parameter	Value	Std. Error	T-value	P(> t)
Emissions as lbs/ton clinker, O₂ as amount				
(Intercept)	8.2	.072	112	$<2 \times 10^{-16}$
poly(TI.550,1)	-28.5	1.16E-03	-30.17	$<2 \times 10^{-16}$
poly(TI.550,2)	-2.4	0.76	-3.1	0.002
O ₂	-6.7E-07	2.00E-06	0.337	0.73
Emissions as lbs/ton clinker, O₂ as on/off				
(Intercept)	8.0	0.11	75	$<2 \times 10^{-16}$
poly(TI.550,1)	-29.6	1.3	-23	$<2 \times 10^{-16}$
poly(TI.550,2)	-2.3	0.76	-3	0.003
Oxygen Enrichment On	0.12	0.136	0.95	0.34
Emissions as lbs/hr, O₂ as amount				
(Intercept)	481	5.8	81	$<2 \times 10^{-16}$
poly(TI.550,1)	-1198	51	-23.2	$<2 \times 10^{-16}$
poly(TI.550,2)	-225	42	-5.4	$1 \times 10^{-8*}$
O ₂	4.2×10^{-4}	1.1×10^{-4}	3.8	2×10^{-16}
Emissions as lbs/hr, O₂ as on/off				
(Intercept)	473	5.8	81	2×10^{-16}
poly(TI.550,1)	-1284	69	81	2×10^{-16}
poly(TI.550,2)	-240	41	-5.9	1×10^{-8}
Oxygen Enrichment On	27	7.4	3.7	2×10^{-4}